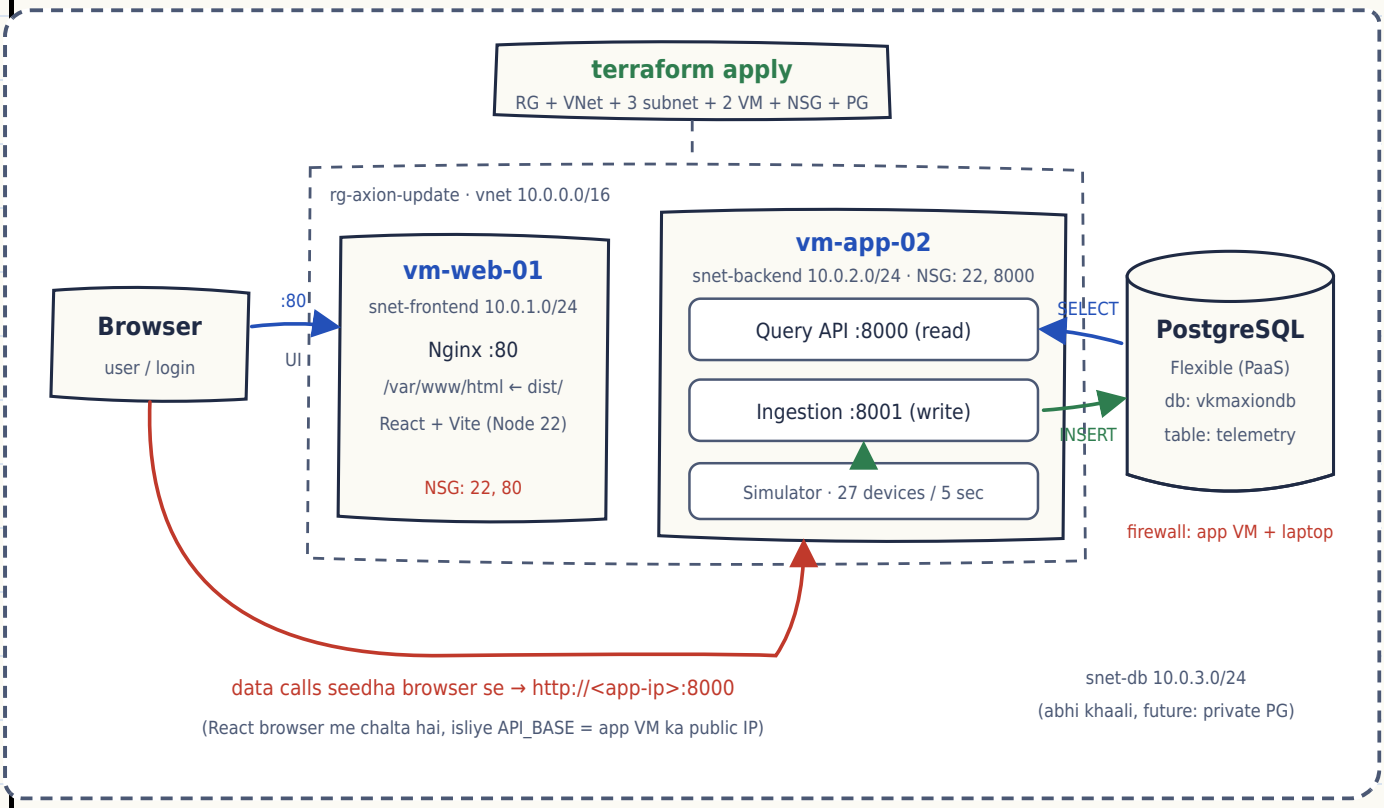
Axion System: 3-Tier on Azure

Zero se poora system dobara banane ka roadmap. Infra Terraform se, DB Azure PostgreSQL, backend 3 Python services, frontend React + Nginx. Upar se neeche, order me chalna.

#0 Kya bana rahe hain



Hara = data likhna (write path). Neela = data padhna (read path). Laal = browser ki API calls.

#1 Samjhane ke liye (simple story)

1 line: Axion ek industrial IoT monitoring app hai. Factory ki machines (pump, motor, compressor) ka temperature, vibration aur current dashboard pe live dikhata hai, aur kharab machines pe alert deta hai.

3 tier:

1. **Presentation:** React dashboard, Nginx serve karta hai (vm-web-01)
2. **Application:** FastAPI services (vm-app-02). Ingestion data *leta* hai, Query data *deta* hai
3. **Data:** Azure PostgreSQL (managed), table `telemetry`

Data ka safar:

1. Simulator har 5 sec me 27 machines ki reading banata hai
2. Ingestion API (`POST /api/v1/telemetry/ingest`) usse DB me INSERT karti hai
3. User browser me dashboard kholta hai, Nginx React files deta hai
4. React browser se Query API (:8000) ko call karta hai, wo DB se SELECT karke JSON deti hai
5. Query API health calculate karti hai: temp > 100 ya vibration > 10 = critical, temp > 85 ya vib > 6 = warning

Interview me bolne wale points

- **IaC:** Terraform modules (RG, VNet, subnet, PIP, VM+NIC+NSG, PG + firewall), remote state Azure Storage me
- **Network security:** har VM ka NSG sirf zaroori port kholta hai. PG sirf whitelisted IPs se accessible, SSL required
- **Read/write split:** ingestion aur query alag services. Load alag scale ho sakta hai (CQRS-type soch)
- **Process management:** sab services systemd pe, `Restart=always` , reboot ke baad khud start
- **Troubleshooting stories:** URL me password encoding, PG firewall ki wajah se timeout, missing table ki wajah se 500 aur React crash, TIMESTAMPTZ vs TIMESTAMP se "Invalid time value"

#2 Roadmap (order mat todna)

A Laptop prep → **B** Terraform apply → **C** DB schema → **D** Query API →
E Ingestion + Simulator → **F** Frontend → **G** Test ✓ → **H** Destroy

kyun? DB table pehle chahiye, warna services 500 dengi. Backend IP pehle chahiye, kyunki frontend build me wahi daalna hai.

A Laptop prep

```
az login
az account show -o table              # sahi subscription?
curl.exe ifconfig.me                  # tera laptop IP
```

Folder: `D:\DevOps Insiders 18\Terraform\axion_server final 01102026\environment\dev\terraform.tfvars` me check kar:

- `my_laptop_ip` = upar wala IP (ghar ka IP badalta rehta hai)
- VMs me `app_port` : web = `"80"` , app = `"8000"`
- `app_vm_public_ip_name = "pip-vm-02"`

B Infra: Terraform

```
cd environment\dev
terraform init
terraform validate
terraform plan        # ~19 to add, 0 to destroy
terraform apply       # PG ko 5-10 min lagte hain
```

Resource group

rg-axion-update (East US)

Network

vnet-axion-landingzone + snet-frontend-01, snet-backend-02, snet-db-03

VMs

vm-web-01 (pip-vm-01) · vm-app-02 (pip-vm-02) · Ubuntu 22.04

PostgreSQL

pgsql-axion-sandbox-vkm-01 · db vkmaxiondb · v16 · Central India

Naye IPs note kar (har baar badlenge)

```
az network public-ip list -g rg-axion-update --query "[].{name:name, ip:ipAddress}" -o table
az postgres flexible-server show -g rg-axion-update -n pgsql-axion-sandbox-vkm-01 --query fullyQualifiedDomainName -o tsv
```

Cheez	Value (yahan likh)
WEB_IP (pip-vm-01)	
APP_IP (pip-vm-02)	

Cheez	Value (yahan likh)
PG_HOST	pgsql-axion-sandbox-vkm-01.postgres.database.azure.com

Dhyan! Agar `apply` bole "name already in use", to purana PG abhi poora delete nahi hua. 5-10 min ruk, ya tfvars me PG ka naam badal de.

C Database: pgAdmin

pgAdmin → Register server → Host: `PG_HOST`, Port 5432, User `vkmadminuser`, Password tfvars wala, SSL mode: **require**. Phir **vkmaxiondb** pe right-click → **Query Tool**.

```
CREATE TABLE telemetry (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    device_id VARCHAR(50) NOT NULL,
    device_type VARCHAR(20) NOT NULL,
    refinery_region VARCHAR(50) NOT NULL,
    timestamp TIMESTAMP NOT NULL,
    temperature DOUBLE PRECISION NOT NULL,
    vibration DOUBLE PRECISION NOT NULL,
    current DOUBLE PRECISION NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
CREATE INDEX idx_telemetry_device_id ON telemetry(device_id);
CREATE INDEX idx_telemetry_timestamp ON telemetry(timestamp DESC);
```

kyun? Ye official schema hai (repo `axion-database-schema`). `TIMESTAMP` bina timezone, kyunki frontend khud end me "Z" jodta hai. `gen_random_uuid()` PG 13+ me built-in hai, isliye pgcrypto extension skip.

D Backend VM: Query API (:8000)

```
ssh vkmadminuser@<APP_IP>
sudo apt update && sudo apt install -y python3-venv python3-pip git
```

DATABASE_URL (teeno services ke liye same)

```
postgresql://vkmadminuser:<PASSWORD>@pgsql-axion-sandbox-vkm-
01.postgres.database.azure.com:5432/vkmaxiondb?sslmode=require
```

Password encoding URL me `@` ko `%40` likhna (jaise `Pass@123` → `Pass%40123`). systemd file me `%` ko `%%`, yaani `%%40`.

```
cd ~
git clone https://github.com/devopsinsiders/axion-telemetry-query-service.git
cd axion-telemetry-query-service
python3 -m venv venv
venv/bin/pip install -r requirements.txt      # fail ho to dobara chala

sudo tee /etc/systemd/system/axion-backend.service > /dev/null <<'EOF'
[Unit]
Description=Axion Query Service
After=network.target

[Service]
User=vkmadminuser
WorkingDirectory=/home/vkmadminuser/axion-telemetry-query-service
Environment="DATABASE_URL=postgresql://vkmadminuser:<PASS-%%40-wala>@pgsql-axion-sandbox-vkm-
01.postgres.database.azure.com:5432/vkmaxiondb?sslmode=require"
ExecStart=/home/vkmadminuser/axion-telemetry-query-service/venv/bin/uvicorn main:app --host
0.0.0.0 --port 8000
Restart=always

[Install]
WantedBy=multi-user.target
EOF

sudo systemctl daemon-reload
sudo systemctl enable --now axion-backend
curl -I localhost:8000/docs          # 200 OK
```

kyun? `--host 0.0.0.0` taaki bahar se (browser se) reachable ho. systemd taaki SSH band hone ya reboot ke baad bhi chalta rahe.

Backend VM: Ingestion (:8001) + Simulator

Ingestion service

```
cd ~
git clone https://github.com/devopsinsiders/axion-ingestion-service.git
cd axion-ingestion-service
python3 -m venv venv
venv/bin/pip install -r requirements.txt

sudo tee /etc/systemd/system/axion-ingestion.service > /dev/null <<'EOF'
[Unit]
Description=Axion Ingestion Service
After=network.target

[Service]
User=vkmadminuser
WorkingDirectory=/home/vkmadminuser/axion-ingestion-service
Environment="DATABASE_URL=postgresql://vkmadminuser:<PASS-%%40-wala>@pgsql-axion-sandbox-vkm-
01.postgres.database.azure.com:5432/vkmaxiondb?sslmode=require"
ExecStart=/home/vkmadminuser/axion-ingestion-service/venv/bin/uvicorn main:app --host 0.0.0.0 --
port 8001
Restart=always

[Install]
WantedBy=multi-user.target
EOF

sudo systemctl daemon-reload
sudo systemctl enable --now axion-ingestion
curl localhost:8001/health           # {"status":"UP"}
```

kyun? Port 8001 isliye kyunki 8000 pe Query API hai. Ye internal hai (simulator usi VM se call karta hai), NSG me kholne ki zaroorat nahi.

Data simulator

```
cd ~
git clone https://github.com/devopsinsiders/axion-data-simulator.git
cd axion-data-simulator
python3 -m venv venv
venv/bin/pip install -r requirements.txt

sudo tee /etc/systemd/system/axion-simulator.service > /dev/null <<'EOF'
[Unit]
Description=Axion Data Simulator
After=network.target axion-ingestion.service

[Service]
User=vkmadminuser
WorkingDirectory=/home/vkmadminuser/axion-data-simulator
Environment="API_URL=http://localhost:8001/api/v1/telemetry/ingest"
Environment="INTERVAL_SECONDS=5"
ExecStart=/home/vkmadminuser/axion-data-simulator/venv/bin/python -u simulator.py
Restart=always

[Install]
WantedBy=multi-user.target
EOF

sudo systemctl daemon-reload
sudo systemctl enable --now axion-simulator
journalctl -u axion-ingestion -f     # har 5 sec "Ingested telemetry" lines
```

kyun? Simulator ka default API_URL Kubernetes wala hai, isliye `localhost:8001` pe point kiya.

F Frontend VM (vm-web-01)

```
ssh vkmadminuser@<WEB_IP>
curl -fsSL https://deb.nodesource.com/setup_22.x | sudo -E bash -
sudo apt install -y nodejs nginx git

git clone https://github.com/devopsinsiders/axion-ui.git && cd axion-ui
npm install

# Backend URL 3 files me hardcoded hai: BUILD SE PEHLE badlo
grep -rln "api.axionsystems.de" src/ | xargs sed -i
"s#https://api.axionsystems.de#http://<APP_IP>:8000#g"
grep -rn "API_BASE =" src/           # teeno me naya URL

npm run build                        # dist/ banega
sudo rm -rf /var/www/html/*
sudo cp -r dist/* /var/www/html/
sudo sed -i 's#try_files $uri $uri/ =404;#try_files $uri $uri/ /index.html;#' /etc/nginx/sites-
available/default
sudo nginx -t && sudo systemctl restart nginx
curl -I http://localhost             # 200 OK
```

kyun? React build ke time URL JS bundle me fix ho jaata hai, isliye baad me badalna ho to dobara build + copy. `try_files` taaki page refresh pe 404 na aaye.

Login (demo, `Login.tsx` me hardcoded): `info@devopsinsiders.com` / `P@ssw01rd@123`

G Test checklist

- ☐ `http://<APP_IP>:8000/docs` → Swagger khulta hai
- ☐ `http://<APP_IP>:8000/devices` → JSON list (500 nahi)
- ☐ pgAdmin: `SELECT COUNT(*), COUNT(DISTINCT device_id) FROM telemetry;` → count badhta hai, devices = 27
- ☐ `http://<WEB_IP>` → login → Fleet Summary me ~27 assets online
- ☐ Device page (chart) khulta hai, black screen nahi
- ☐ `sudo reboot` ke baad bhi sab services chalti hain

```
systemctl status axion-backend axion-ingestion axion-simulator --no-pager
journalctl -u axion-backend -n 30 --no-pager
```

H Kaam khatam: Destroy

```
cd environment\dev
terraform destroy
```

kyun? VMs aur PG chalte rahe to paisa katta rehta hai. State remote storage me hai, to agli baar seedha Phase A se shuru.

#3 Errors + fix

Error / symptom	Matlab	Fix
`pip: no matching distribution`	Slow network	pip install dobara
`ConnectionRefusedError [Errno 111]`	DATABASE_URL set nahi, localhost pe ja raha	systemd/export me DATABASE_URL
`Weird host / URL parse error`	Password ka `@`	`%40` (systemd me `%%40`)
`asyncio TimeoutError`	PG firewall block	tfvars `my_laptop_ip` / app VM rule check, `nc -zv PG_HOST 5432`
`API 500 + e.reduce is not a function`	telemetry table nahi	Phase C, phir `systemctl restart axion-backend`

Error / symptom	Matlab	Fix
`RangeError: Invalid time value`	Column TIMESTAMPTZ hai	Official schema (TIMESTAMP)
`Dashboard 0 assets online`	Simulator / ingestion band	`systemctl status axion-simulator axion-ingestion`
`Page refresh pe 404`	Nginx try_files nahi laga	Phase F ka sed
`-bash: rg: No such file`	`<placeholder>` copy ho gaya	Asli value daal
`az: command not found (VM)`	VM pe Azure CLI nahi	Laptop se chala
`Map: API KEY REQUIRED`	Map tiles key nahi	Ignore

#4 Next level (todo)

- Passwords tfvars se hata ke Key Vault / `TF_VAR_` env vars me
- PG private access (snet-db-03 delegate + private DNS), public access band
- Nginx reverse proxy `/api` → 8000, port 8000 bahar se band
- cloud-init / Ansible se VM setup automate
- Docker images (repos me Dockerfile hai) → AKS + ArgoCD

~ Vishal · Axion lab · Oct 2026